DRAFT

Sandbags for Home Runoff

Fill sandbags efficiently using a multi-bag filling fixture.

BEFORE YOU BEGIN

Sandbags can redirect shallow stormwater runoff from vulnerable areas of your property. They work best when placed before heavy rain arrives.

WHAT YOU'LL NEED

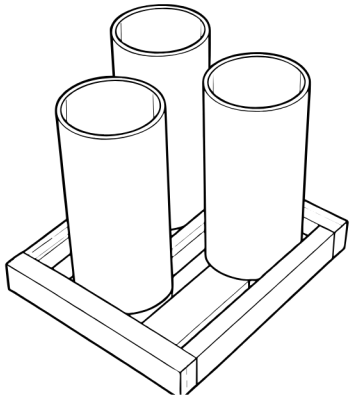
- Empty woven sandbags
- Clean sand or suitable granular fill
- Multi-bag filling fixture
- Shovel
- Gloves and sturdy footwear
- Eye protection when sand is dusty
- A clear, level work area
- A partner for lifting

△ CAUTION — PREVENT HARM

A filled bag may weigh 35–40 lb. Use a partner or lighter bags if needed. Work on level, dry ground; keep hands and feet clear when lifting the fixture. Leave bags one-half to two-thirds full, with fabric to fold underneath; remove excess sand.

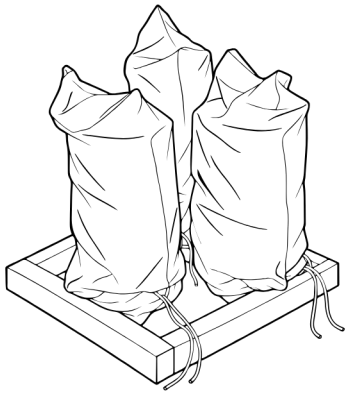
FILL SANDBAGS WITH THE FIXTURE

1 **Invert the fixture.**



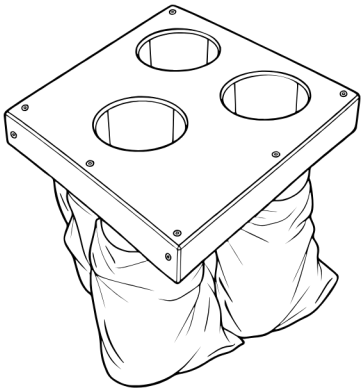
1. Invert the fixture.

2 **Put a sandbag on each pipe.**



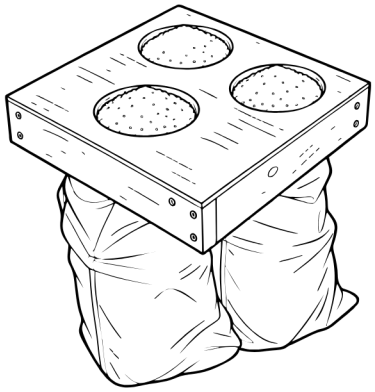
2. Put a sandbag on each pipe.

3 **Turn the fixture right side up.**



3. Turn the fixture right side up.

4 **Fill the fixture openings with sand.**



4. Fill to the top with sand.

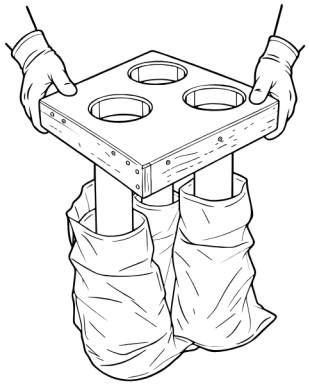
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Sandbags for Home Runoff — continued

Remove filled bags, shape them for placement, and store them safely.

COMPLETE THE FILLING CYCLE

5 **Lift the fixture and leave the sand in the bags.** Use a partner and lift together.



5. Lift straight up. Let the filled bags remain on the ground.

6 **Set the fixture upside down.**



6. Reset the fixture for more sandbags.

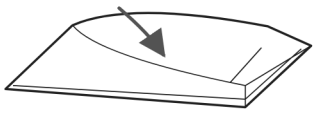
7 **Move the sandbags away.**



Four sandbags lying in a row.

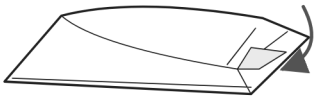
SHAPE AND FOLD THE BAGS

8 **Flatten sand toward the closed end.**



8. Flatten the sand toward the closed end.

9 **Fold the empty end underneath.** Do not tie barrier bags.



9. Fold the empty end underneath.

STORE FILLED SANDBAGS

- Stack low on a pallet or stable support.
- Protect from sun and weather.
- Keep driveways and exits clear.
- Inspect for damage before use.



Sandbags should rotate 90-degrees at each layer to interleave like bricks with the bags below

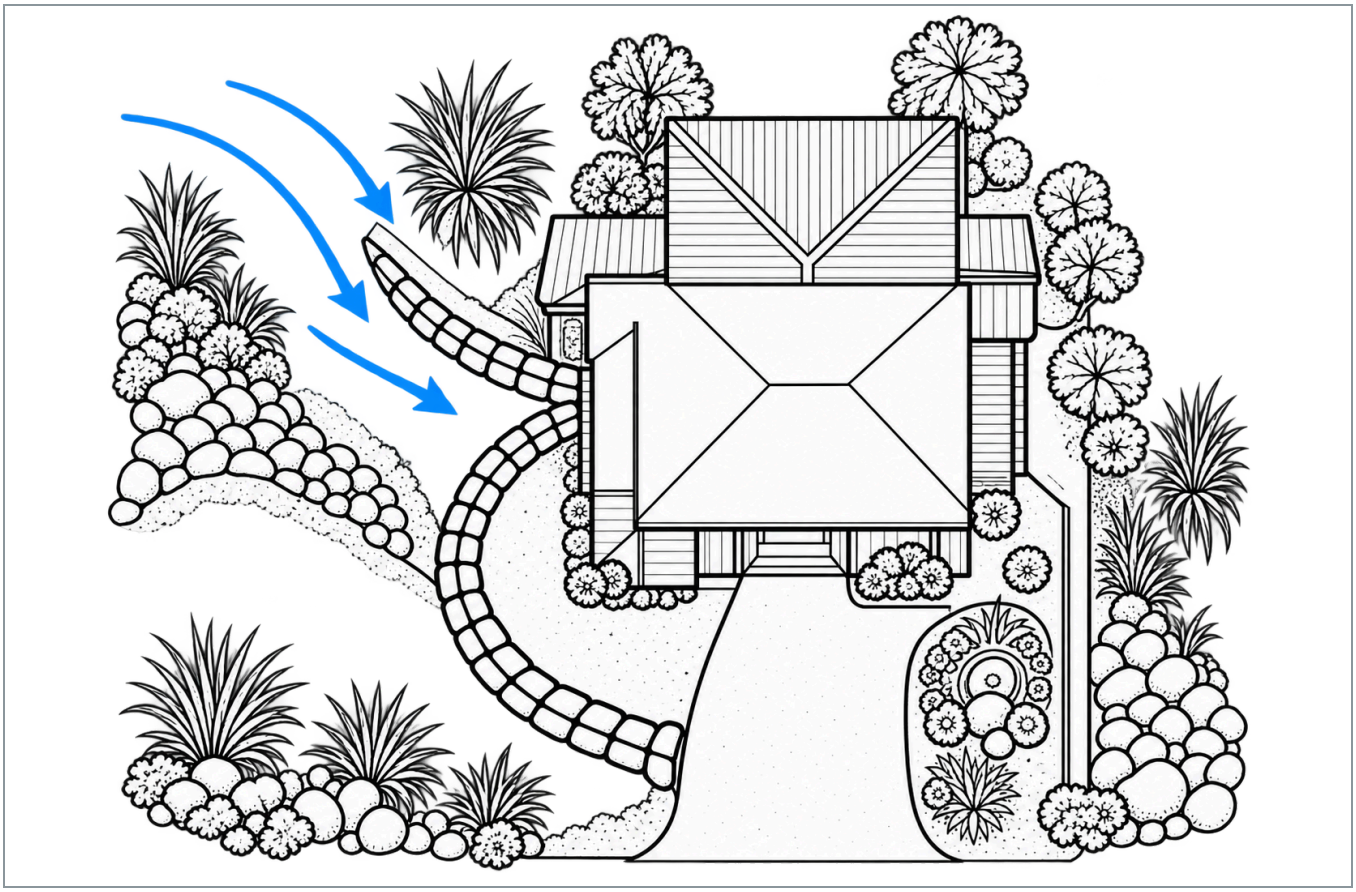
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Planning a Sandbag Barrier

Understand where water comes from and where it can safely go.

FIRST, OBSERVE THE WATER'S PATH

1. **Where is the water coming from?** Look for runoff from hillsides, driveways, roofs, and low areas.
2. **What needs protection?** Identify vulnerable doors, garages, low windows, and openings.
3. **Where can the water safely go?** Find an existing drainage path on your property.



Use sandbags to redirect water to a safe outlet. Do not dump your water into your neighbors lawn.

SELECT THE BARRIER LOCATION

- Place bags on firm, level ground.
- Intercept runoff before it reaches the opening.
- Use a short line that redirects water.
- Keep drains, exits, and access routes open.
- Do not redirect runoff toward a neighbor, road, or another doorway.

× DANGER — STOP NOW

Leave if water is moving fast, rising around you, or cutting off your exit. Do not enter flowing water or build beside an unstable slope or wall.

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Building a Sandbag Barrier

Overlap bags, stagger joints, and maintain a stable base.

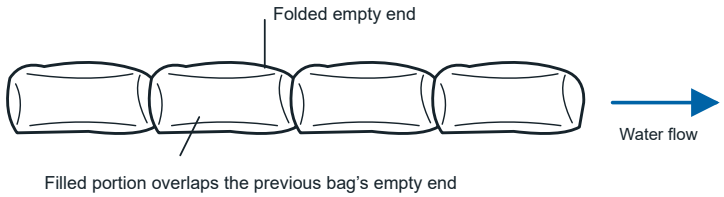
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PREPARE THE GROUND

1. **Clear the area.** Remove loose debris and fill low spots.
2. **Establish the line.** Identify the water-facing side and intended direction of flow.

PLACE THE FIRST COURSE

3. **Lay bags flat.** Fold the empty end underneath and toward incoming water.
4. **Overlap adjoining bags.** Lay the filled part over the previous bag's empty end.
5. **Seat the bags.** Press or tamp firmly to reduce gaps.



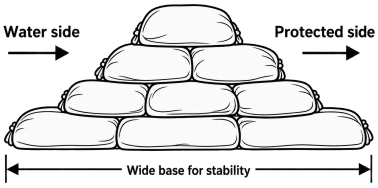
Top view — overlap adjoining bags.

ADD ADDITIONAL COURSES

6. **Stagger every course.** Offset joints by about half a bag.
7. **Maintain a wide base.** Keep the barrier low with a stable, stepped profile.
8. **Check the ends.** Connect to suitable higher ground or provide a safe outlet.



Sandbags staggered around a basement window.



As you increase the height of a dyke, increase the width of the dyke as well.

INSPECT THE FINISHED BARRIER

- Check for gaps, uncovered joints, and displaced bags.
- Look for water flowing around the ends or pooling behind the line.
- Sandbags are temporary and do not form a watertight seal.

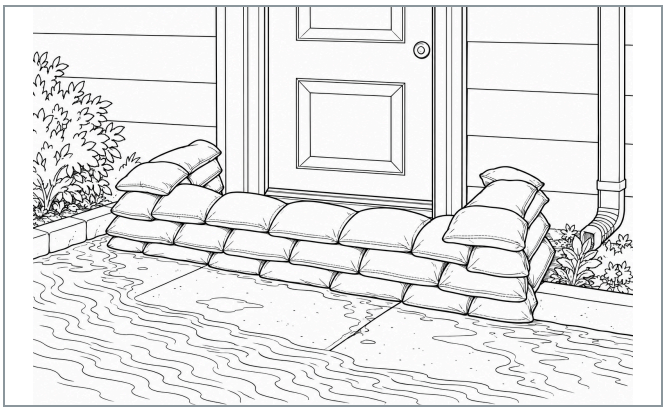
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Protecting Vulnerable Openings

Choose the barrier arrangement that fits the runoff problem.

A. EXTERIOR DOORWAYS

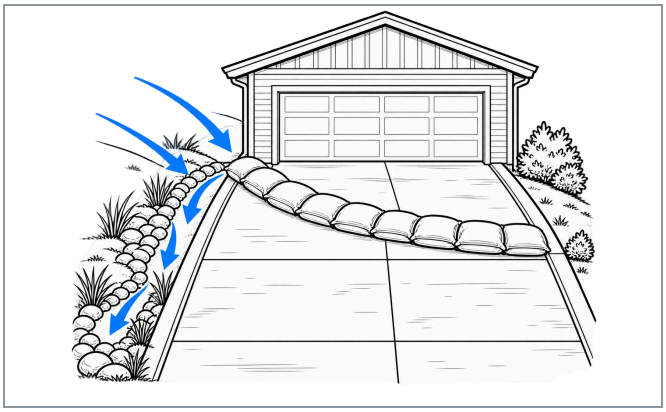
- Place a short, low barrier before water reaches the threshold.
- Extend it beyond both sides of the opening.
- Avoid blocking the only safe exit.
- Check for water entering around the ends or through other openings.
- Do not rely on sandbags to make a doorway watertight.



Doorway protection. Keep another safe exit available.

B. GARAGE DOORS AND SLOPING DRIVEWAYS

- Intercept runoff uphill of the garage door.
- Use an angled or curved line to redirect water.
- Lead water to an existing suitable drainage path.
- Avoid directing water toward a neighbor, road, or other doorway.
- Keep the evacuation route open.

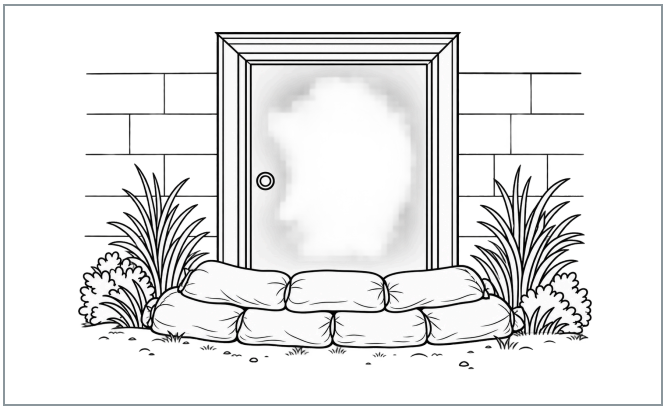


Use sandbags to divert water flow along safe paths. If it overtops the sandbags, give it a safe location for the water to flow.

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C. LOW WINDOWS AND UTILITY OPENINGS

- Identify low windows, window wells, and utility penetrations.
- Use a low diversion barrier to reduce surface runoff.
- Keep required ventilation, window escape routes, and utility access clear.



If you place sandbags around a door, give yourself enough room to enter and exit.

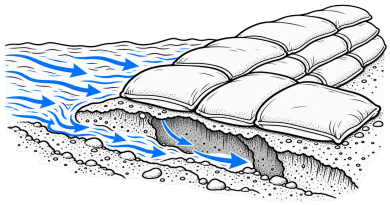
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Understanding Water Movement

Recognize bypass, seepage, and unintended pooling.

1. WATER FLOWING AROUND THE ENDS

A line that stops below higher ground may send water around its end.



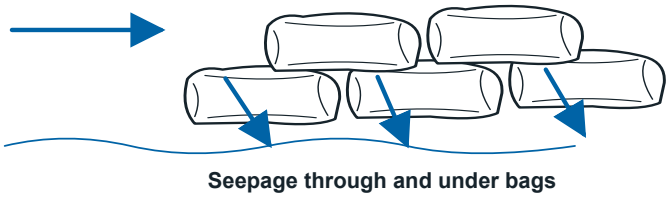
Fast moving water is undermining a sandbag installation

What to do

- Extend or adjust the line only from safe, dry ground.
- Use suitable higher ground or an existing safe outlet.

2. WATER SEEPING THROUGH THE BARRIER

Water can pass between bags, through fabric, and underneath.



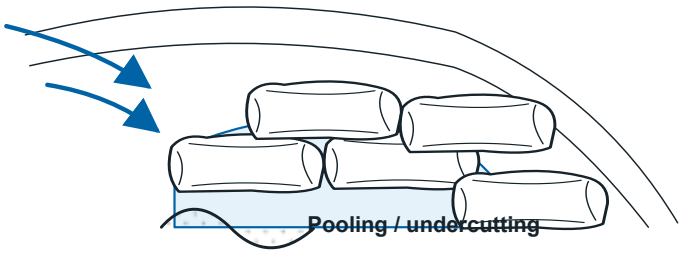
Water seeping through the barrier.

What to do

- Seat bags firmly and cover joints.
- Follow local guidance for water-side plastic sheeting.
- Inspect for leakage from safe ground.

3. WATER POOLING BEHIND THE BARRIER

A poorly placed barrier can trap water on the protected side.



Water pooling behind the barrier.

What to do

- Do not work in flowing water or on unstable ground.
- Move to safety if erosion, displacement, or a rapid rise occurs.

× DANGER — STOP NOW

Leave if conditions become unsafe. Undercutting, displaced bags, or rising water can cause the barrier to fail.

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Sandbag Mistakes to Avoid

Six common errors that reduce protection.

1. Aligned vertical joints

× WRONG



Aligned joints

✓ RIGHT

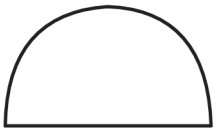


Offset joints

bag joints line up vertically. the upper bag bridges the joint below.

2. Overfilled bags

× WRONG



Overfilled and rounded

✓ RIGHT



Partly filled and flexible

an overfilled rounded bag cannot settle tightly. a partly filled bag lies flat and remains flexible.

3. Tall, narrow stacks

× WRONG



Tall, narrow base

✓ RIGHT

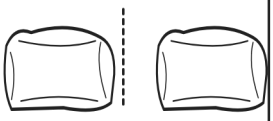


Wide, stepped base

four bags form a narrow column. a low stepped stack has a broad stable base.

4. Open gaps and loose ends

× WRONG



Gaps and loose ends

✓ RIGHT

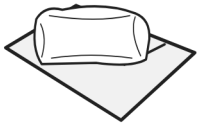


Overlap and fold tails

separated bags leave a gap. flattened bags overlap with empty ends folded underneath.

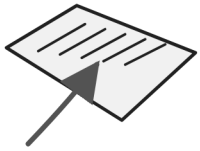
5. Blocking drainage

× WRONG



Drain blocked

✓ RIGHT

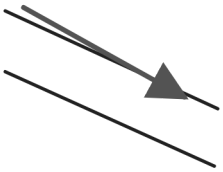


Keep drains open

a sandbag blocks a drain grate. the drain is clear, with a labeled runoff arrow entering it.

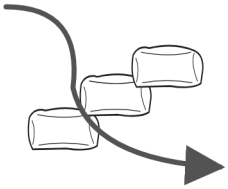
6. Unsafe diversion

× WRONG



Toward neighbor / road

✓ RIGHT



To a safe outlet

runoff arrows point toward a neighbor or road. a low line steers water to an existing safe outlet on the property.

× DANGER — STOP NOW

Keep exits open. Do not work in moving water or build beside an unstable wall or slope. Leave if conditions become unsafe.

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Sandbag Field Checklist

Prepare early, inspect safely, and remove temporary barriers when conditions permit.

BEFORE THE STORM

- ☐ Identify vulnerable openings and runoff paths.
- ☐ Locate a safe, functioning drainage route.
- ☐ Clear accessible drains and drainage channels.
- ☐ Obtain sandbags, sand, and necessary tools.
- ☐ Fill and stage bags on stable ground.
- ☐ Build the diversion before water arrives.
- ☐ Check that exits remain accessible.

DURING THE STORM

- ☐ Observe runoff from a safe location.
- ☐ Watch for water bypassing the barrier.
- ☐ Check for unexpected pooling or seepage.
- ☐ Stay away from moving water and unstable slopes.
- ☐ Stop work and leave if conditions become dangerous.



Inspect from safe ground; keep exits open.

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AFTER THE STORM

- ☐ Wait until conditions are safe before approaching.
- ☐ Inspect for erosion, unstable ground, and displaced bags.
- ☐ Remove temporary barriers when no longer needed.
- ☐ Do not dump used sand into drains or natural channels.
- ☐ Follow local disposal guidance for wet or contaminated bags.
- ☐ Correct drainage problems before the next storm.

△ CAUTION — PREVENT HARM

Sandbags redirect and slow water; they do not make a home floodproof. Do not enter floodwater, approach unstable slopes, or stay in a threatened structure to maintain a barrier.

SOURCES

U.S. Army Corps of Engineers, St. Paul District. *Flood Fight Handbook* (2022), §§2.1–2.3. Flood Fight Handbook PDF.
RiverOakRanch. “Filling Sandbags the Easy Way, With Sandbag Tubes” (2010), Instructables. Fixture instructions. Both sources accessed 2026-09-24.